

Tab 1

Exprimment -1

Experiment–1: Linux Programming Environment & Basic Commands

A) Setting up of Linux Programming Environment:

Aim:

To study and practice basic Linux commands for terminal operations, file and directory management, file permissions, and input/output redirection in the Linux programming environment.

Requirements / Software Used

Kali Linux / Linux OS

Terminal

vi editor.

Theory / Introduction:

Linux kya hai, terminal/shell kya hota hai, aur Linux commands ka basic purpose — around 4–6 lines.

Linux Programming Environment

C programming can be performed either online or offline according to our requirements. For programming, it is useful to work in a Linux environment.

1. Online Method

CoCalc.ai:

CoCalc is an online platform used to write and run C programs.

Steps:

Open cocals in a web browser.

Sign in or create an account.

Create a new project.

Open the terminal.

Create a C program file.

Write the C program.

Compile and run the program using the terminal.

2. Offline Methods

C programming can be done in a Linux environment using the following methods:

a) Dual Boot:

In dual boot, two operating systems are installed on the same computer. We can select any one operating system when the computer starts.

b) Single Boot:

In single boot, only one operating system is installed and used on the computer.

c) Bootable Pen drive:

A bootable pen drive contains a Linux operating system. Linux can be started or installed using the pen drive.

d) Virtual Machine:

A virtual machine allows us to run Linux inside another operating system without installing Linux directly on the computer.

1) Virtual Machine Manager (Hypervisor)

A: hypervisor is software that creates and manages virtual machines.

Type-1 Hypervisor (Bare Metal)

A Type-1 hypervisor runs directly on the computer's hardware. It does not require a normal operating system as a host.

Examples: VMware ESXi, Microsoft Hyper-V.

Type-2 Hypervisor (Hosted)

A Type-2 hypervisor runs on an existing operating system and is used to create and run virtual machines.

Examples: Oracle Virtual Box and VMware Workstation.

For this practical, we will use Oracle Virtual Box, which is a Type-2 (Hosted) hypervisor.

2) Installation of VMM & VM

Step 1: Download Oracle Virtual Box for Windows.

Step 2: Download the pre-built Kali Linux VM for Virtual Box.

Step 3: Extract the downloaded Kali Linux VM file.

Step 4: Install Oracle Virtual Box on the computer.

Step 5: Open Oracle Virtual Box.

Step 6: Open the saved Kali Linux VM in Virtual Box.

Step 7: Start the Kali Linux VM.

Step 8: Open the terminal in Kali Linux. Create a simple C program, then compile and run it to understand how the Linux programming environment works.

Result:

Thus, the installation of Type-2 VMM (Oracle Virtual Box) and VM (Kali Linux) was completed successfully.

B) Working with Linux Commands we can execute the commands below in your Kali Linux terminal one section at a time.

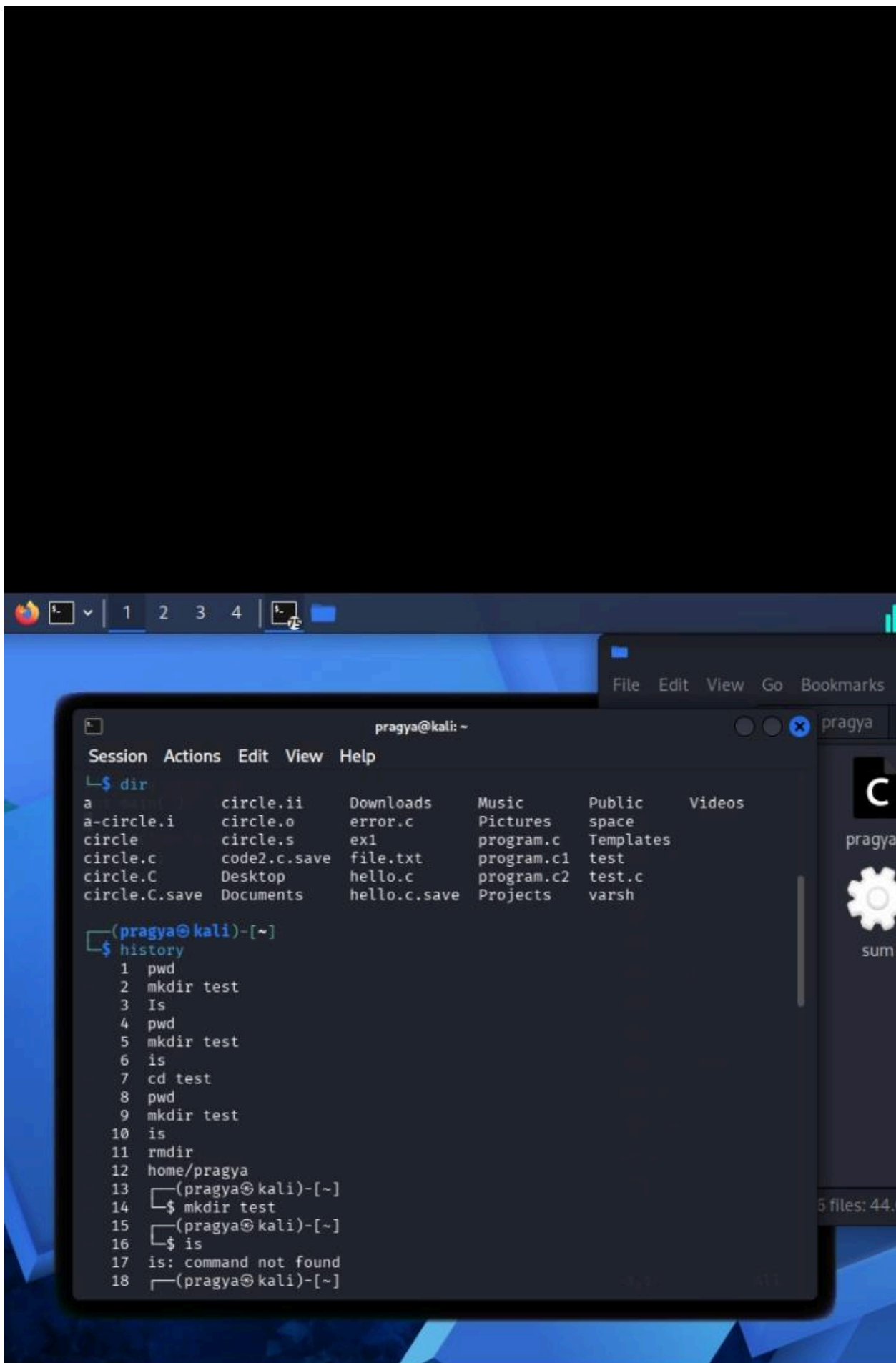
Terminal basics: pwd, mkdir, rmdir, dir, ls, cd, cd .., clear, history, man, whoami.

Commands + Explanation :

Terminal basics: pwd, mkdir, rmdir, dir, ls, cd, cd .., clear, history, man, whoami

Files: vi, file, rm, cp, mv, cat, more, less, head, tail, chmod

Redirection: >, >>, <



1 2 3 4

```
pragya@kali: ~  
Session Actions Edit View Help  
(pragya@kali)-[~]  
$ whoami  
pragya  
(pragya@kali)-[~]  
$ mkdir test  
mkdir: cannot create directory 'test': File exists  
(pragya@kali)-[~]  
$ ls  
a circle circle.c circle.C circle.C.save circle.ii circle.o circle.s code2.c.save Desktop Documents Downloads error.c ex1 file.txt hello.c hello.c.save Music Pictures program.c program.c1 program.c2 Projects Public space Templates test test.c varsh Videos  
(pragya@kali)-[~]  
$ dir  
a circle circle.c circle.C circle.C.save circle.ii circle.o circle.s code2.c.save Desktop Documents Downloads error.c ex1 file.txt hello.c hello.c.save Music Pictures program.c program.c1 program.c2 Projects Public space Templates test test.c varsh Videos  
(pragya@kali)-[~]  
$
```

Files: vi, file, rm, cp, mv, cat, more, less, head, tail, chmod

Redirection: >, >>, >

1) pwd

Displays the current working directory.

2) mkdir

Creates a new directory.

3) rmdir

Removes an empty directory.

4) dir

Displays files and directories.

5) ls

Lists directory contents.

6) cd

Changes the current directory.

7) cd ..

Moves to the parent directory.

8) clear

Clears the terminal screen.

9) history

Shows previously executed commands.

10) man

Displays the manual/help page for a command.

11) whoami

Displays the current logged-in username.

12) vi

Opens the vi text editor to create or edit a file.

13) file

Identifies the type of a file.

14) rm

Removes a file.

15) cp

Copies a file or directory.

16) mv

Moves or renames a file.

17) cat

Displays file contents.

18) more

Displays file contents one screen at a time.

19) less

Allows scrolling through file contents.

20) head

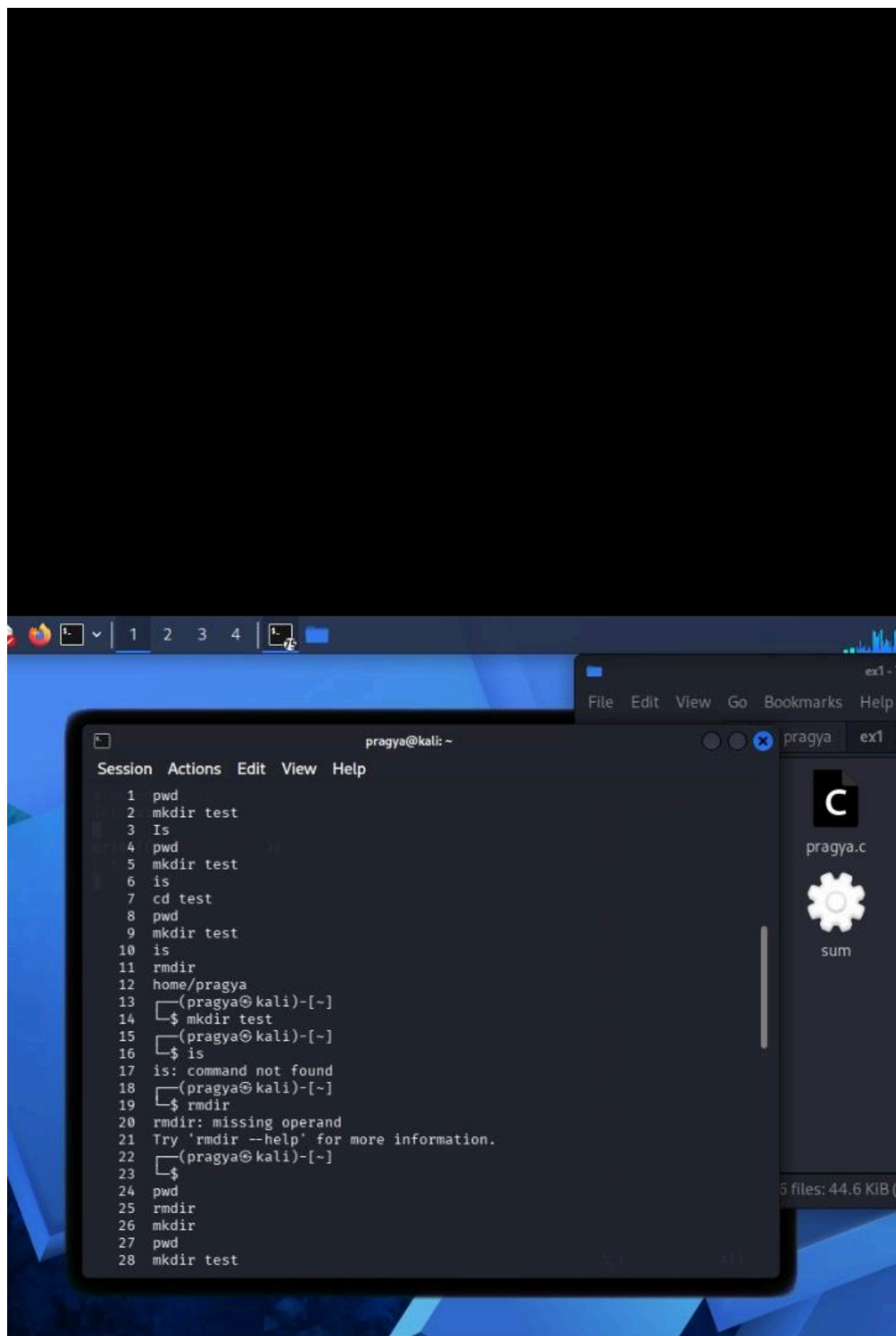
Displays the beginning of a file.

21) tail

Displays the end of a file.

22) chmod

Changes file permissions.



pragya@kali: ~
Session Actions Edit View Help

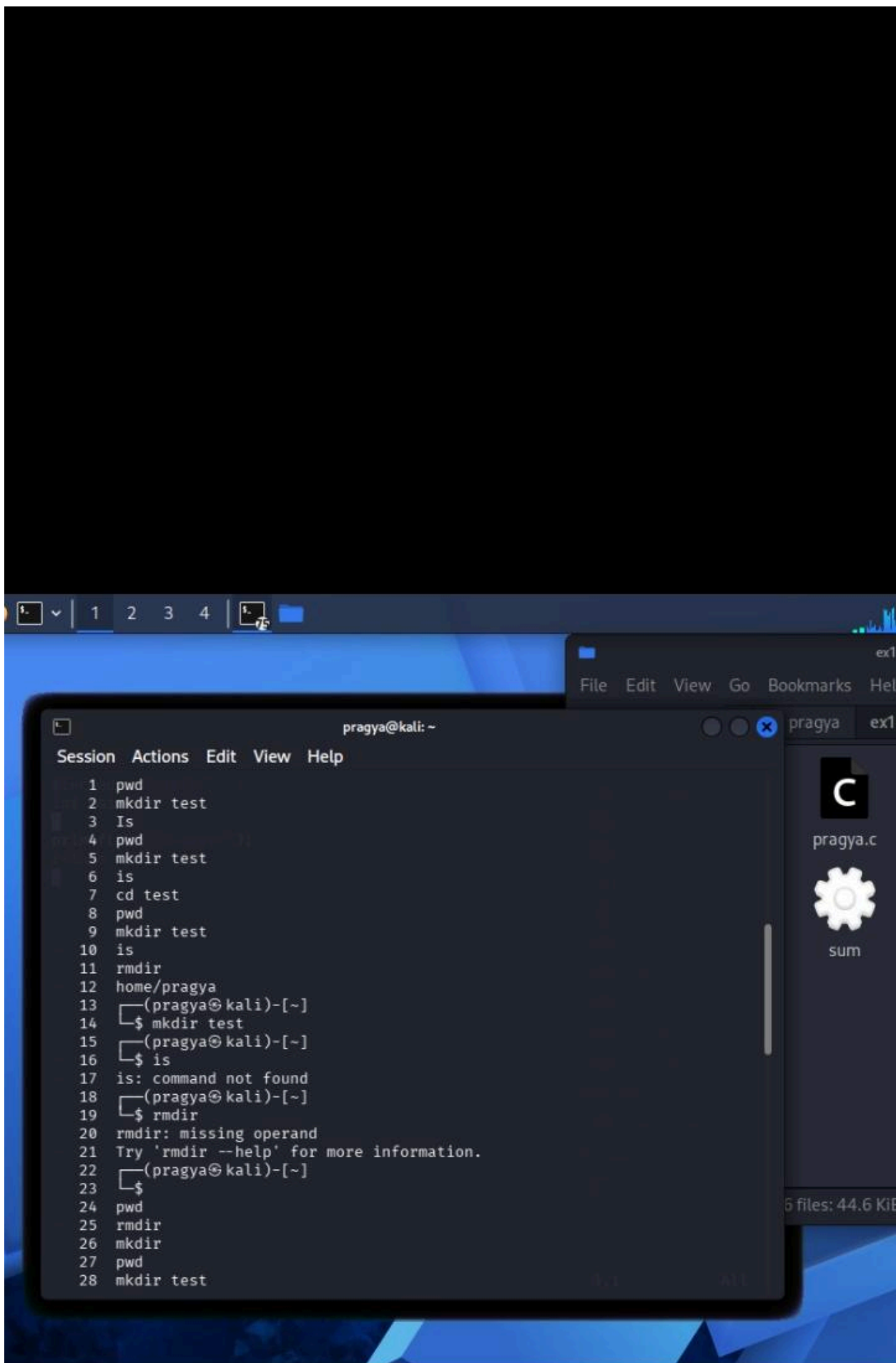
```
1 pwd
2 mkdir test
3 ls
4 pwd
5 mkdir test
6 ls
7 cd test
8 pwd
9 mkdir test
10 ls
11 rmdir
12 home/pragya
13 (pragya@kali)-[~]
14 └─$ mkdir test
15 (pragya@kali)-[~]
16 └─$ ls
17 ls: command not found
18 (pragya@kali)-[~]
19 └─$ rmdir
20 rmdir: missing operand
21 Try 'rmdir --help' for more information.
22 (pragya@kali)-[~]
23 └─$
24 pwd
25 rmdir
26 mkdir
27 pwd
28 mkdir test
```

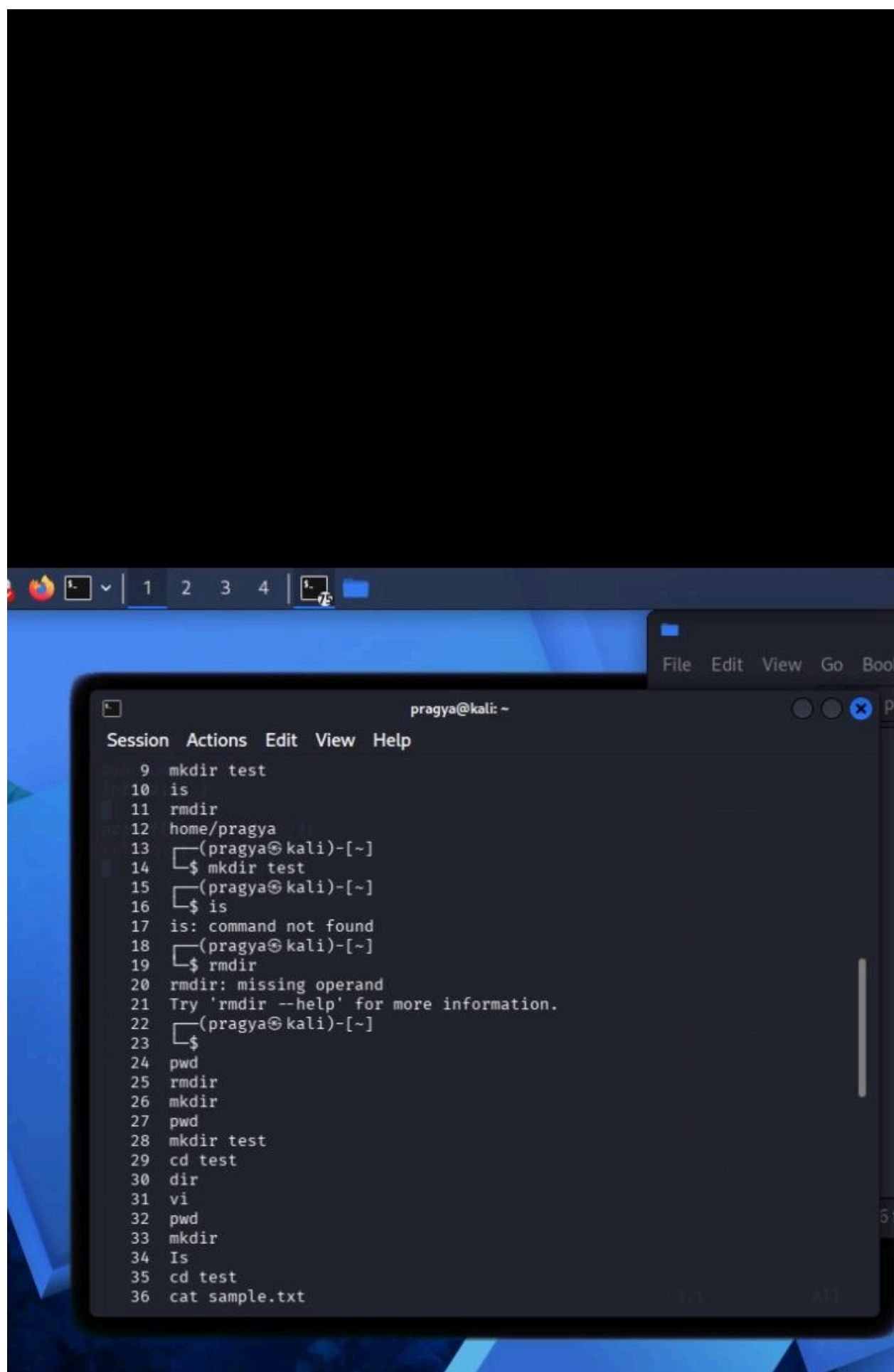
File Edit View Go Bookmarks Help

pragya ex1

pragya.c
sum

5 files: 44.6 KIB





```
pragya@kali: ~
Session Actions Edit View Help
9 mkdir test
10 is
11 rmdir
12 home/pragya
13 (pragya@kali)-[~]
14 $ mkdir test
15 (pragya@kali)-[~]
16 $ is
17 is: command not found
18 (pragya@kali)-[~]
19 $ rmdir
20 rmdir: missing operand
21 Try 'rmdir --help' for more information.
22 (pragya@kali)-[~]
23 $
24 pwd
25 rmdir
26 mkdir
27 pwd
28 mkdir test
29 cd test
30 dir
31 vi
32 pwd
33 mkdir
34 Is
35 cd test
36 cat sample.txt
```

Result / Conclusion:

Result: Basic Linux commands for directory management, file operations, permissions, and input/output redirection were successfully executed and studied.

Thank you.

Name: Pragya Pandey

Course : btech(cse)

Sap id:590034352

Tab 2

